

EMT Power System Modeling Seminar

Human Resource Development for Future Power System Planning and Operation – **2nd Session**



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22nd January 2026

Key Points from Seminar 1

- **Modern Power Systems with High Penetration of Inverter-based Resources (IBRs)**
 - IBRs have introduced new types of stability problems to the connected network, such as sub-synchronous oscillations (SSO) and converter-driven stability.
- **Phasor Domain Transients (PDT) vs Electromagnetic Transients (EMT) software**
 - New IBRs introduced stability phenomena cannot be studied in PDT domain due to limited bandwidth, therefore EMT software is adopted.
- **Building EMT Study Capabilities**
 - Transmission system operators (TSOs) worldwide have started to build EMT study capabilities in-house. However, there are challenges to be tackled.

Seminar 2 Contents

- **State-of-the-art in EMT**
 - Overview of EMT study capability development by various Transmission System Operators (TSOs) worldwide, including motivations, lessons learned and incoming developments.
- **EMT Modelling Requirements**
 - EMT modelling requirements in grid codes, including functional requirements, model details and test cases etc.
- **Large-scale EMT case studies**
 - Examples of TSOs carrying out large-scale EMT studies.

List of Transmission System Operators Investigated

Europe

- EirGrid (TSO in Ireland)
- Energinet (TSO in Denmark)
- Fingrid (TSO in Finland)
- NESO (TSO in GB)
- TenneT (TSO in Germany and Netherland)

Australia

- AEMO (Australia Energy Market Operator)

North America (NA)

- ERCOT (Electric Reliability Council of Texas)
- ISO-NE (Independent System Operator – New England)
- MISO (Midcontinent System Operator)
- Southern Company (Southeastern System Operator)



State-of-the-art in EMT Study Capability Development

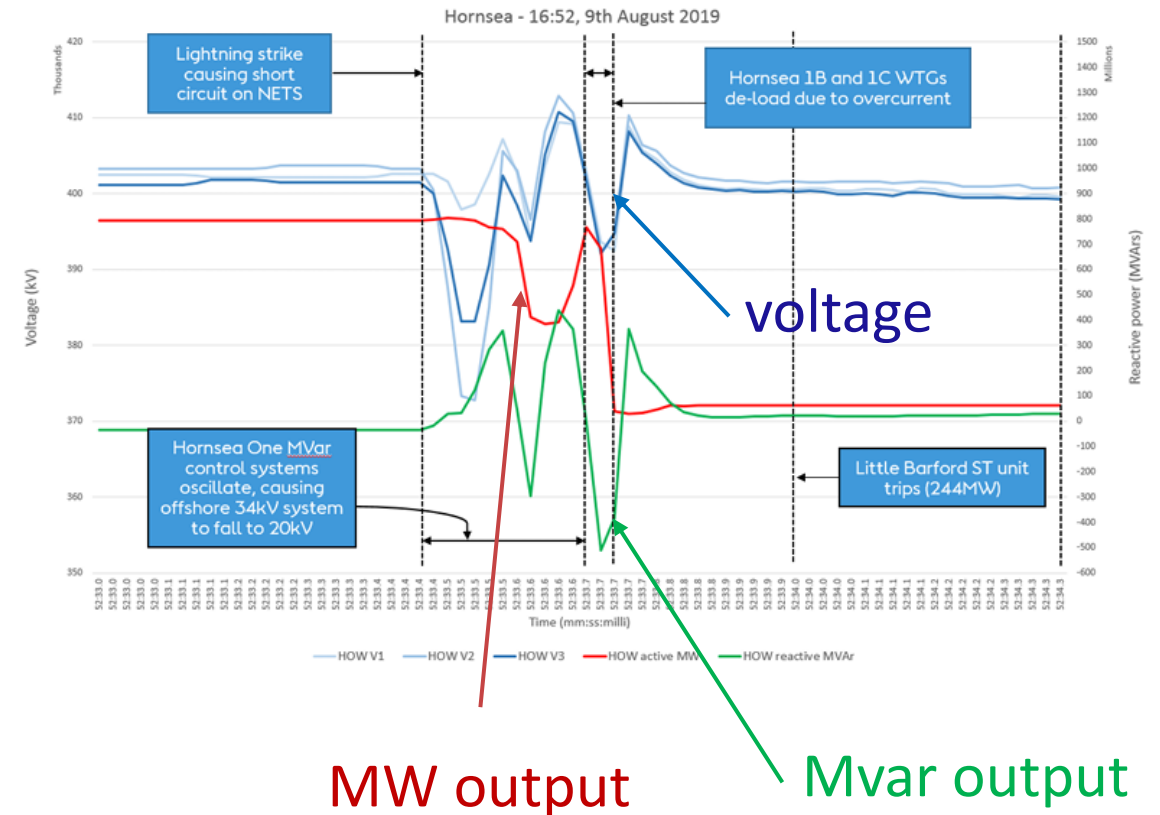
Main Motivations for EMT Studies

- All TSOs started process due to **increased awareness of IBRs' impact on their electricity transmission systems**, with few TSOs also already observed IBR related system instability issues (e.g. NESO system event in August 2019, - see next slide).
- TSOs observed that **IBRs response cannot be fully captured using PDT software** while there is a need to identify possible dynamic issues in planning time frame. For HVDC application, dynamic performance studies in EMT domain was one of the earliest applications experienced by TSOs. Hence the need to replicate this for IBRs also.
- TSOs have identified **multiple critical use cases for EMT studies** including connection of IBRs, large-scale stability studies and post-event investigations etc.

NESO System Event – 9th August 2019

- The event was initiated by a **lightning strike** resulting in the **disconnection of a 400kV circuit**.
- A large offshore windfarm went into an underdamped oscillatory mode with **reactive power fluctuating between +400 and -560 Mvar**.
- 730MW of the windfarm's output was de-loaded by overcurrent protection.
- The event led to several unexpected and cascading events resulting in approx. **1.7GW generation and 1 GW demand disconnection**.

Hornsea Offshore Windfarm Output during system event on 9th Aug 2019



Graph from the report "Orsted technical Report for National Grid ESO on the events of 9 August 2019".

State-of-the-art in EMT Study Capability Building

In four key areas

■ Grid Code EMT Modelling Requirements

- All TSOs investigated have EMT modelling requirements for SG and IBRs (wind, solar, HVDC and BESS) power plants. At high level, [the requirements are similar but can differ in detail](#). Some TSOs are working on EMT requirements for large load and IBRs with grid forming capabilities. This is an ongoing task.

■ Large-scale EMT studies

- Few TSOs have built their entire or most parts of their transmission network in EMT domain and carry out large-scale EMT studies for investigating system events or for network planning purposes.

■ IT & Computing Infrastructure Upgrade

- TSOs have upgraded their IT and computing infrastructure for EMT studies. The upgrade will [continue with increasing complexity of EMT studies](#). Some TSOs have plan to utilize cloud-based server to carry out large-scale EMT studies.

■ Recruitments

- TSOs have expanded their resource pool ranging from [5 to 30+ EMT study engineers/experts depending on the size of the networks and frequency of EMT studies](#). The recruitment is also on going. Some TSOs also starts to build an EMT data management team.

Leanings from Developing EMT Modelling Requirements in Grid Code

- It is important to have an **early engagement** between TSO and other relevant parties including the regulatory authority, original equipment manufacturers (OEMs), existing and prospective network users and subject-matter experts.
- The following modelling aspects should be discussed and clearly defined:
 - functional requirements
 - compliance tests and accuracy requirements
 - software compatibility
 - submission documentation
 - maintenance and support
- Need for **a panel consisting of representatives from all stakeholders** to have regular meetings to discuss, review and propose changes to the EMT requirements in grid code.

Learnings from Improving EMT Study Skills in-house

- The quickest way to improve EMT study capability is through **recruitments**. Target recruitment candidates include those with Master or PhD qualifications with relevant academic training, and EMT experts from consultant companies or OEMs.
- As EMT study is a relatively specialized area, it can take long time to train junior/graduate engineers. However, TSOs found it can be **effective to train up engineers who already have good experience in dynamic studies in PDT domain**.
- Some TSOs have used third party consultants to provide EMT training. However, the training is normally relatively basic. It is more effective and necessary to **train engineers on the job**, i.e., by working directly on EMT modelling and studies of TSOs' network.

Learnings from Upgrading IT and Computing Infrastructure

- EMT studies demand high computation power to reduce simulation times, therefore, investing in additional IT & computing infrastructure is a priority..
- Procuring high computing workstations can have long lead time and lack of scalability due to space constraints etc.
- Cloud-based approach can be highly scalable and flexible; however, simulation speed can be uncertain.
- See a typical example of computing infrastructure effects on speed of simulations in the next slide.

ISO-NE Example: Cloud-based EMT Simulation Pilot Project

- ISO-NE carried out a pilot project to evaluate cloud-based simulations.
- Same test cases with increasing number of IBR connections (1,2,4,6,12,24) were tested on:
 - local premium mini workstation
 - local HPC dual CPU server
 - cloud-based server
- Results with 24 IBR connections indicate that cloud-based and HPC server outperform the local mini workstation. It is expected that the **cloud-based will outperform local HPC server once the number of IBRs on the test system increases further.**

Key Challenges Encountered

- It remains a major challenge to **acquire EMT models of existing/legacy plants**, due to either the EMT models not being available or compatibility issues between available models and the existing EMT software platforms. Some TSOs tried to develop generic EMT models for these plants but parametrization appears to be very challenging.
- There is a **wide variation in the software versions/compilers provided by OEMs**. This is an issue for TSOs when integrating these models within a wide-area system level model.
- **Lack of dedicated IT human resources and/or understand of computing needs from existing IT resources** has hindered the progress of IT and computing infrastructure upgrade for EMT studies.
- Recruiting EMT experts can be challenging due to scarcity of resources in the market.

Example: Building EMT Model of GB Transmission Network

utilizing innovation fund

- To overcome the lack of modelling limitations in PDT simulation tools in studying weak, low inertia systems, an [innovation project “Transmission Tools for EMT Modelling” \(TOTEM\)](#) was kicked off by the three Transmission Owners (TOs) in GB (NGET, SP and SSE) and NESO to build the whole GB transmission network in EMT domain.
- Expectation was that EMT based models will address these concerns and seen as an enabler of de-risking the integration of the power electronics-based technologies.
- Objectives of the initiative:
 - Build and validate a EMT model of the GB transmission network;
 - Provide a path for the GB TOs to adopt the models and put them into use; and
 - Establish shared computing resources for running the most complex models.
- The project have been [extended to TOTEM 2 \(completed\)](#) and [TOTEM 3](#) started since June 2025 to continuously improving the EMT model in following aspects:
 - enhance simulation speed
 - develop UK-standard key protection model
 - improve network conversion process from PDT to EMT

Building EMT Study Capability – The Next Stage

- Depending on the current progress, TSOs have identified the following future work:
 - the need for developing robust screening methods to identify when complex EMT study is required (see Fingrid example in the next slide)
 - move to use of standardized Dynamic Link Library (DLL) in model provision to address software version and compiler issues
 - expanding EMT network models and performing system wide EMT studies
 - introducing EMT modelling requirements for large load
 - building dedicated EMT model management team
 - developing possible use cases of EMT studies for real-time operations
 - upgrading IT & computing infrastructure and weighing pros and cons between local and cloud based EMT studies

Fingrid Example: Applying EqSCR Screening Method

- Fingrid has conducted EMT studies as part of their outage planning process in addition to load flow and dynamic studies in PDT.
- Equivalent Short Circuit Ratio (EqSCR) is adopted to determine whether large-EMT studies should be carried out.
- If $\text{EqSCR} < 2$, i.e., risk of IBRs converter instability is regarded as high leading to EMT studies.
- In Fingrid, around 5% of outage planning cases required EMT studies.

$$\text{EqSCR}_i = \frac{S_i}{P_i + \sum \left(\frac{\Delta V_j}{\Delta V_i} + P_j \right)}$$

considering the impact of other IBRs

S_i = fault level at bus i

P_i = maximum MW of IBR connected at bus i

P_j = maximum power of IBR connected at bus j

$\frac{\Delta V_j}{\Delta V_i}$ = sensitivity of voltage at bus j due to voltage change at bus i



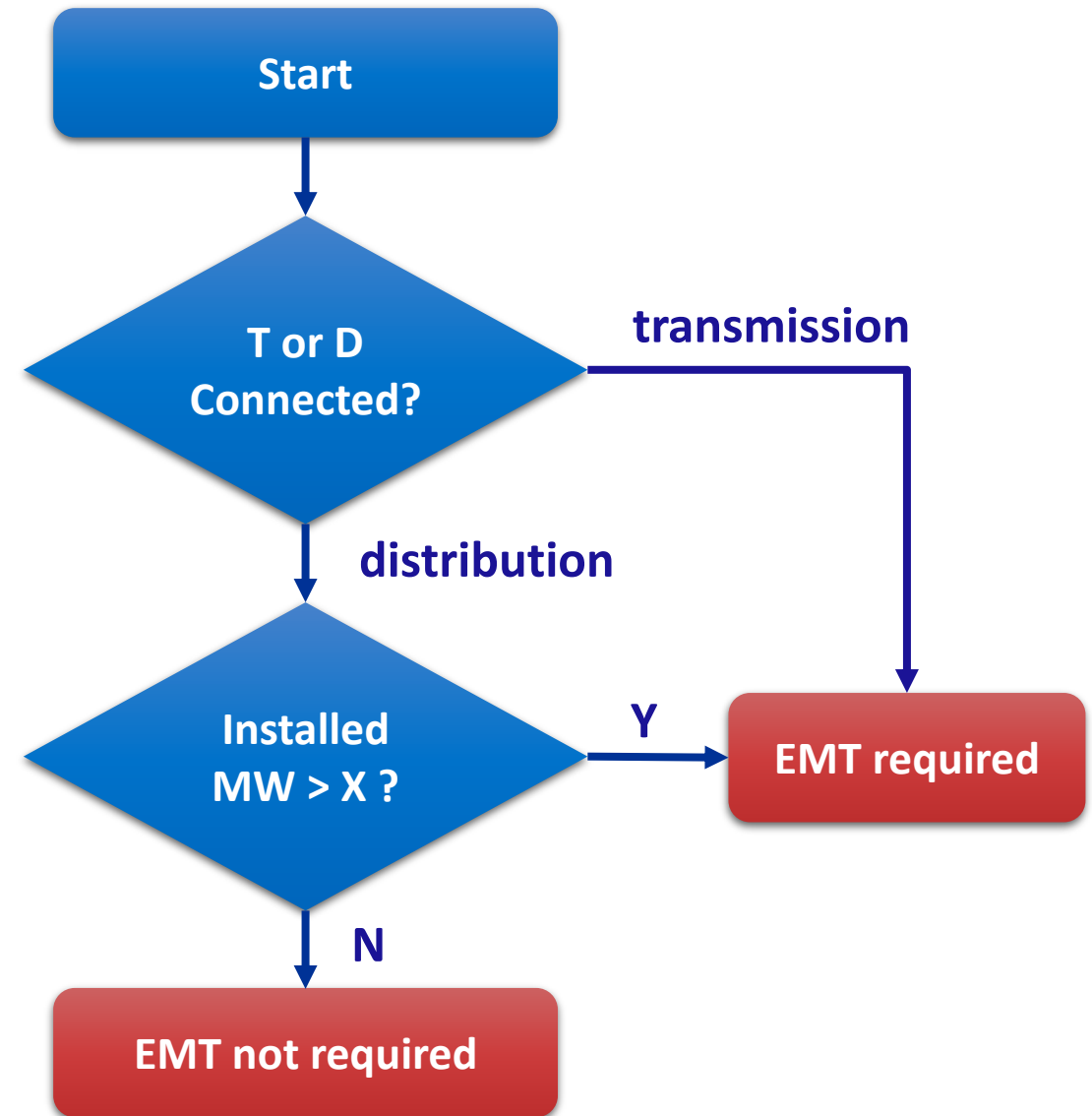
Grid Code EMT Modelling Requirements

Main Coverages in Grid Code for EMT Modelling Requirements

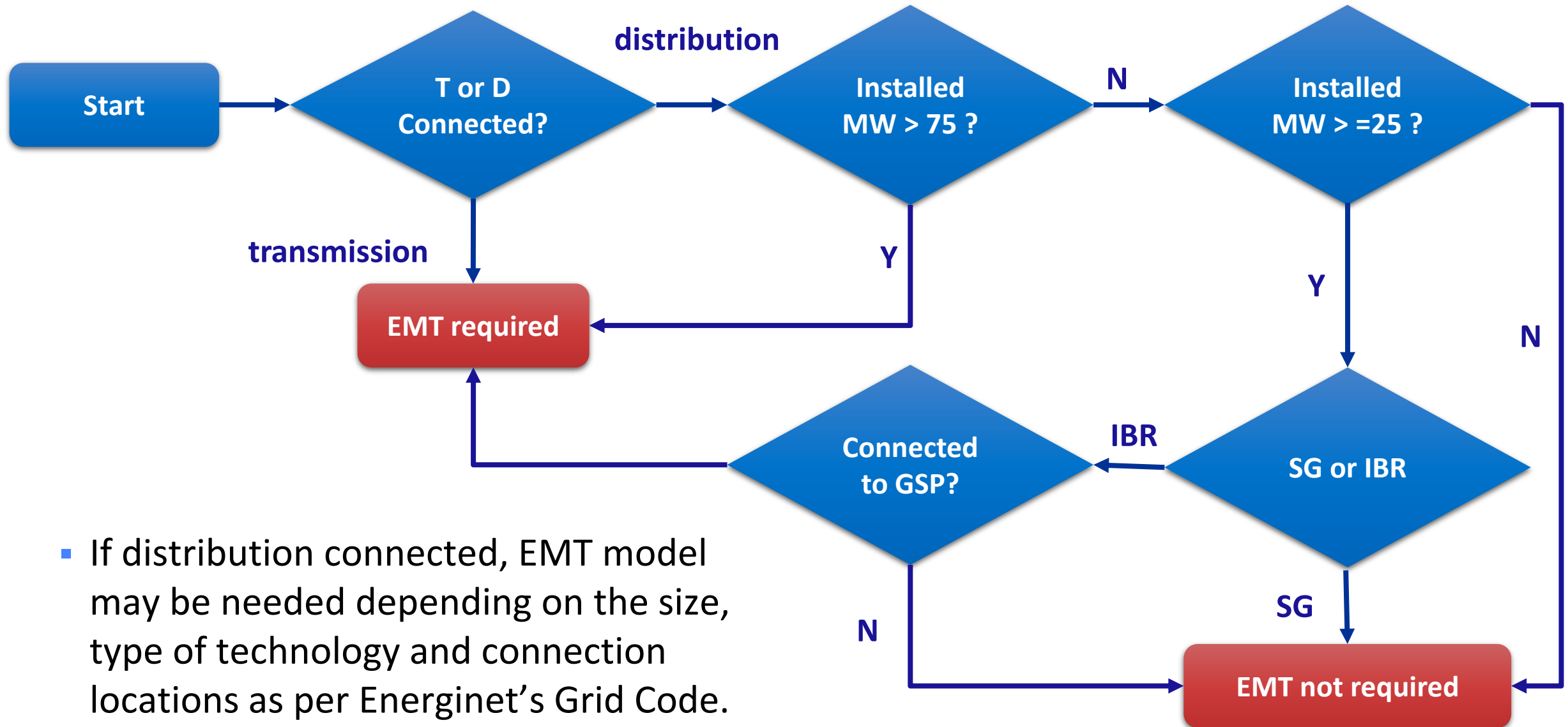
- Who shall provide EMT models?
- List of intended uses/applications.
- What components should be modelled?
- Model compliance tests and accuracy requirements.
- Usability requirements.
- Model submission requirements.
- Model maintenance and support requirements.

Provision of EMT models – Who?

- Flow chart shows general filtering process; however, variations exist between different TSOs.
- The filtering depends on **types of power plant, transmission or distribution connected** and the **maximum MW capacity**.
- EMT models are required if **the user system is transmission connected**. If distribution connected, models may be required depending the plant size.



Energinet example: EMT model provision



- If distribution connected, EMT model may be needed depending on the size, type of technology and connection locations as per Energinet's Grid Code.

Intended use of EMT models - What for?

- TSOs define list of studies in which EMT models will be used, including:
 - classical power system stability studies
 - convertor-driven stability and SSO studies
 - harmonic studies
 - small-signal disturbances (set-point changes)
 - transformer energization studies
 - Fault-ride-through (FRT) studies
 - protection performance studies
 - black-start or islanding operation studies (if applicable)

Modelling of Components

- Plant models required for:
 - SG
 - IBRs
 - wider user system components
 - protection
- Models should cater for specific input/output requirements.
- The EMT models shall produce **similar response compared to real plants installed on-site** (TSOs do not accept generic models for new plants). All the parameters and settings should be site-specific.
- The EMT models must be sufficiently detailed to have **frequency bandwidth from [0, 0.2] to [2, 2.5] kHz**.

Synchronous Generator Components Modelling Requirements

- Electrical representation of synchronous machine, excitor with its control, AVR, PSS, turbine and governor control.
- Whole mechanical drive-train represented as **multi-mass shaft torsional model**, including inertia constant, shaft spring constants, torques share and damping of each drive-train mass.

Required EMT turbine generator shaft model parameters by Fingrid

Parameter	Unit
Inertia Constant	kgm ²
Shaft Spring Constant	Nm/rad
Turbine Torque Share	P.U.
Self Damping	Nms/rad
Mutual Damping	Nms/rad
Speed Variation Damping Factor	P.U.

Table is from Fingrid “Modelling instruction for PSS/E and PSCAD models” published in 2024.

IBR Component Modelling Requirements

- For wind generation, electrical generator, including aerodynamic components, mechanical drive-train.
- For battery energy storage system (BESS), power storage unit.
- Converter, including DC link etc. Converter switching dynamics can be either represented as either **switching or average model**.
- Full representation of Power Plant Controller (PPC), inner and outer loop control, Phase-Locked Loop (PLL).
- All relevant control functions including **FRT, P, Q, voltage, pitch control (wind), battery management system (BESS), any special controls** such as grid forming control and power oscillation damping control etc.

Wider User System Component Modelling Requirements

- Any other equipment as part of the whole user system, including:
 - harmonic filters and reactive compensation devices
 - circuit breakers
 - surge arrestors
 - cables and overhead lines
 - transformers with magnetizing branch and tap changer details
 - auxiliaries

- Controls to be included:
 - transformer tap changer control
 - reactive compensation device control (both static and dynamic plant details)

Protection Modelling Requirements

- All relevant protection functions must be modelled including:
 - over/under frequency, rate of change of frequency (ROCOF)
 - over/under voltage
 - overcurrent protection
 - harmonic protection
 - for SG, loss of excitation, out-of-step, torsional stress protection, excitation limiter etc.
 - protection relevant to FRT
 - negative sequence protection
 - for IBRs, SSO protection (if applicable)
- For most TSOs, all protection functions including filtering process must be based on the **‘real-code’** as defined in standard IEEE/CIGRE DLL vendor agnostic format, as defined by CIGRE TB 958.

EMT Model Input/Output Requirements

- Adjustable user input:
 - control setpoints, such as P , Q , f , V
 - control modes, such as P - f , Q - V droop, fixed P , fixed Q , V control, Power Factor control etc.
 - control parameters, such as ramp rates, deadbands, limits
 - all relevant protection settings, including enable/disable protection
- Measurable signals:
 - P , Q , V , I , f at connection point and generator terminal
 - for IBR: PLL output, internal control reference for measuring voltage and currents in d-q domain, DC link voltages etc.
 - for SG: rotor angle and speed, inputs/outputs of AVR, PSS, excitation system, governor
 - protection signals that can cause tripping, activation of FRT etc.

EMT Model Simulation Tests – Europe

- All TSOs in Europe requires the following key simulation tests:
 - **Real and Reactive Power Setpoint Step Changes:** demonstrate the user system model represent the plant dynamic performance by applying P, Q, voltage and power factor reference step change.
 - **Limited Frequency Sensitive Mode – Over/Under-frequency (LFSM-O/U):** demonstrate compliance MW response to increase/decrease in system frequency while operating in LFSM mode.
 - **Frequency Sensitive Mode (FSM):** demonstrate compliant MW response to changing system frequency while operating in FSM mode.
 - **Fault Ride Through and Fast Fault Current Injection:** demonstrate capability of fault ride through and voltage dips.
- Additional tests can be required by different TSOs, such as high voltage right through (HVRT), phase angle jump, protection, power oscillation damping control etc.

TenneT example: LFSM-O Simulation Test

- TenneT required LFSM-O to be triggered once frequency increase beyond 50.2Hz.
- The table indicates LFSM-O step change tests set out by TenneT with 5% droop.
- Test is successful if:
 - MW change meets the level indicated in the table
 - within acceptable accuracy tolerance when comparing to the measured value
 - no underdamped oscillations
- In addition to frequency step change tests, frequency ramp tests are also required.

Applicable frequency step change required by TenneT

Starting frequency (Hz)	To frequency (Hz)	Power change per step
50.0	50.2	0%
50.2	50.5	-12%
50.5	51.0	-20%
51.0	51.5	-20%
51.5	51.0	+20%
51.0	50.5	+20%
50.5	50.2	+12%
50.2	50.0	0%

LFSM-O test points corresponding to 5% droop

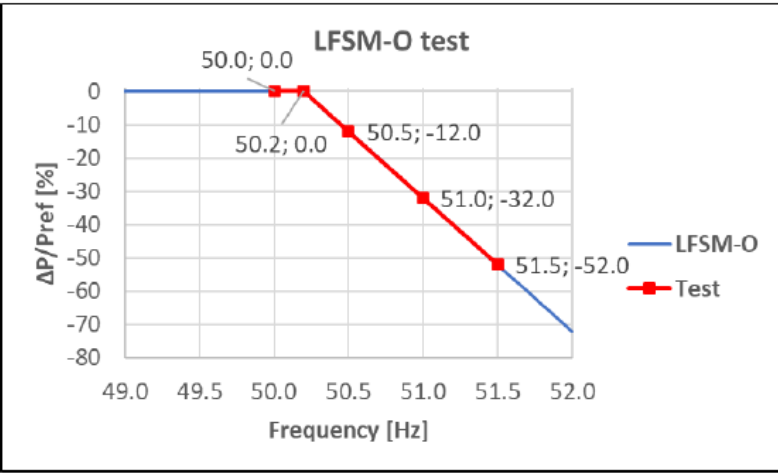


Table and Figure from TenneT “Power-Generating Modules compliance verification”, 2024.

EMT Model Simulation Tests – North America

- Types of EMT verification tests required by TSOs in North America generally in-line with those in Europe, however details can differ.
- Requirements for EMT model test cases are referenced to IEEE 2800-022.

Southern Company

1. Initialization (Flat start) test
2. Balanced fault ride-through tests
3. Small voltage disturbance test
4. Small frequency disturbance test
5. Voltage or Power Factor (V/PF) step response tests
6. Active power step response tests
7. High Voltage Ride-Through (HVRT) tests
8. Low Voltage Ride-Through (LVRT) tests
9. High-Frequency Ride-Through (HFRT) tests
10. Low-Frequency Ride-Through (LFRT) tests
11. Short Circuit Ratio (SCR) tests
12. Protection verification tests
13. Reactive power capability tests

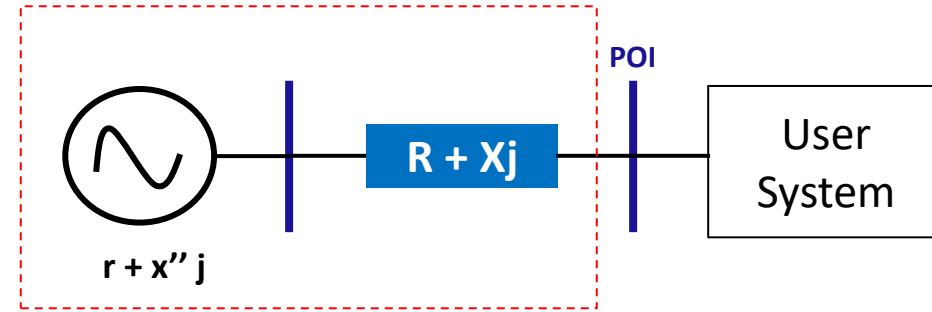
MISO

1. Initialization (Flat start) test
2. Balanced fault ride-through tests
3. Small voltage disturbance test
4. Small frequency disturbance test
5. High Voltage Ride-Through (HVRT) tests
6. Low Voltage Ride-Through (LVRT) tests
7. High-Frequency Ride-Through (HFRT) tests
8. Low-Frequency Ride-Through (LFRT) tests
9. Protection verification tests
10. Short Circuit Ratio (SCR) tests
11. Phase Angle Change tests

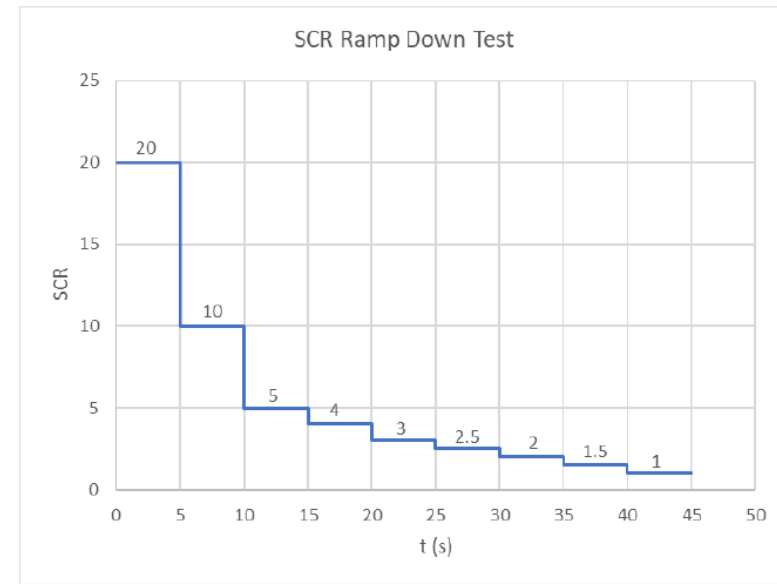
MISO Example: Short Circuit Ratio Simulation Test

- MISO requires testing IBR model stability by **changing SCR from 20 to 1** at point of interconnection (POI) shown in the figure.
- The initial MW of IBR is set to the maximum, and Q set to 0 Mvar at POI.
- The plant shall not trip and have **stable and well-damped response with $SCR \geq 2.5$, with acceptable damping ratio ≥ 0.3** .
- MISO expects grid-forming IBR has even lower SCR stability threshold (E.g. 1.2 by ERCOT).

Single Machine Infinite Bus (SMIB) Test Network



MISO SCR Simulation Test



Graph from Electranix "PSCAD-Model-Requirements-Rev.-13", 2025

EMT Model Accuracy Requirements

- Some EMT simulation test results are required to compare to real test results, while some need to compare to results from PDT simulations.
- In general, EMT model response can be divided into three phases:
 - **Pre-event period:** the time immediately before the start a disturbance event.
 - **Transient period:** The time between the start of the disturbance and the time when the model's response settles within a predefined tolerance (e.g. 5%) of the final steady state value.
 - **Post-event period:** the time starting from **x** seconds (e.g. 1s) after the transient period is passed.
- EMT model response is compared to the user system measured response in each phase with different accuracy requirements set by TSO.

NESO example: EMT Model Accuracy Requirements

- Pre-event period accuracy criteria:
 - simulated value should be within $\pm 3\%$ of measured value

- Transient period accuracy criteria:
 - for 95% of the time within the transient period, the difference between simulated response and measured response shall be contained within an envelope defined by a $\pm 10\%$ deviation around the simulated response
 - simulated response shall have similar trend the measured response. E.g. if oscillation observed in simulated response but not in measured response, then the accuracy requirement is not met

- Post-event period accuracy criteria:
 - simulated value should be within $\pm 3\%$ of measured value

Requirements for EMT Model Usability

- Simulation time steps should be agreed with TSO. Different TSOs have different requirements but typically between 10-20 μ s. ERCOT requires minimum timestep to be 1 μ s, but most TSOs expect 10 μ s should be the minimum that the EMT model can produce accurate response for all studies considered.
- Any limitations on model performance while running on specific time steps shall be notified.
- TSOs require EMT model shall be initialized and reach steady state within 3-5s of simulation time, more compiled model such as HVDC is allowed to take longer, e.g. 6s by NESO. The model is required to be numerically stable for at least 20 (e.g. ERCOT) -100 (e.g. Eirgrid) seconds.

Requirements for EMT Model Submissions

- EMT model file shall include all components in the whole user system.

- A User Manual including detailed descriptions of the followings:
 - model version
 - all components and their parameter values, control block diagrams
 - limitations of the model, such as SCR, X/R, time steps etc
 - all user adjustable inputs and outputs
 - how to set up and initialize of the model
 - all the relevant AC and DC protections and settings accessible to the user
 - verification report

EMT Model Maintenance and Support Requirements

- Model providers should confirm they have model maintenance and support frameworks in place with vendors [through the lifetime of the equipment](#).
- Technical support from the model supplier must be made available to support the TSO to solve any issues related to the model.
- Updated dynamics data for an existing plant shall be provided to when field tests, inspections, or other information demonstrates that the dynamics data should be changed to accurately represent the dynamic characteristics of the plant.
- The owner of a dynamic element connected to the transmission system shall verify the dynamic models through comparison with operational data of actual events or test results on a regular basis after energization.

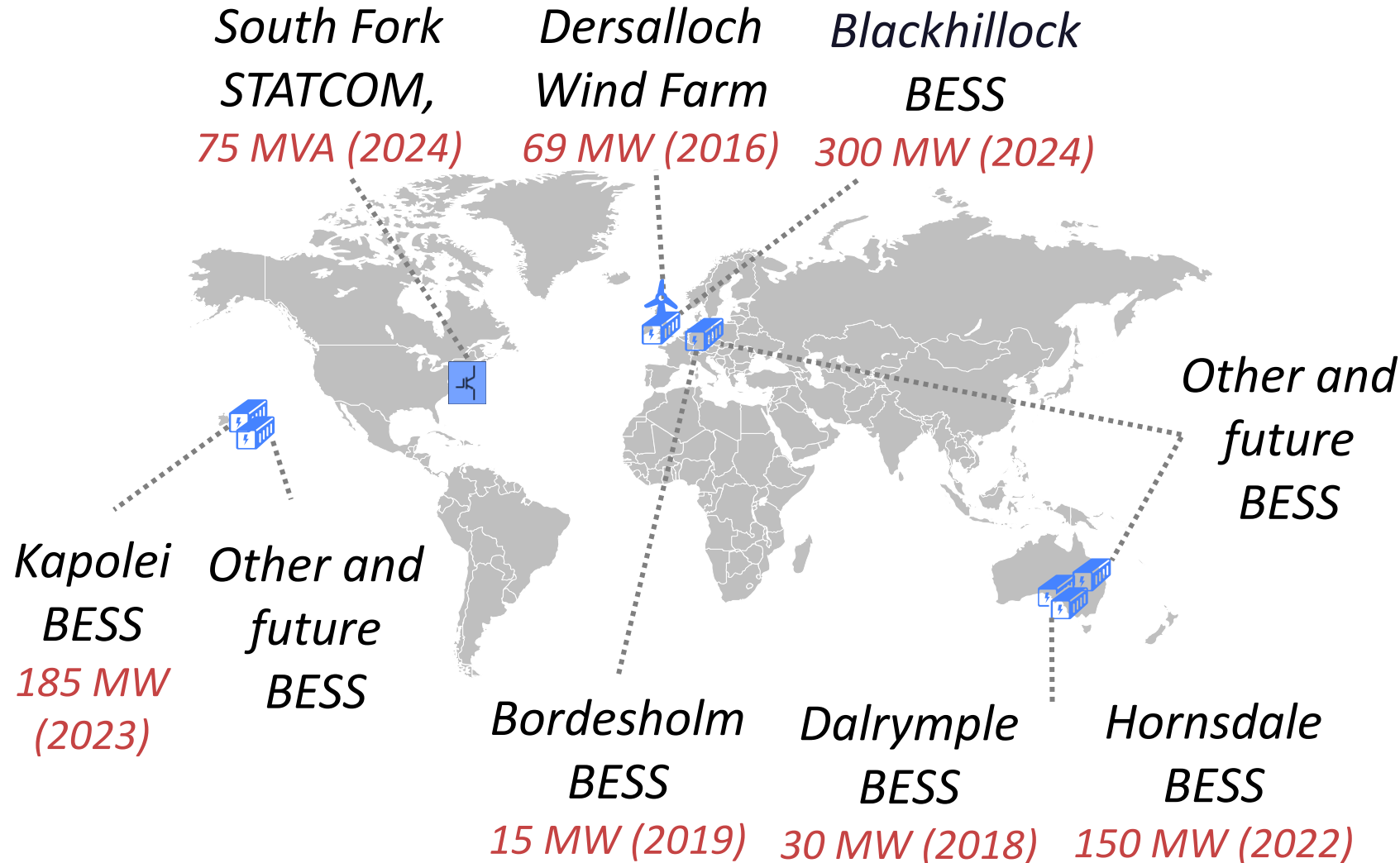
Comparing High-level EMT Model Requirements

Category	EMT Model Requirement	ERCOT	ISO-NE	MISO	Southern Company	AEMO
Model Accuracy Features	Converter bridge model	Y	Y	Y	Y	Y
	Fully represent inner control loops	Y	Y	Y	Y	Y
	Include all pertinent control features	Y	Y	Y	Y	N
	Include plant-level controls	Y	Y	Y	Y	N
	Include plant grounding	N	N	N	N	N
	Utilize actual hardware code *OR* validate the model	Y	N	Y	Y	N
	Include electrical and mechanical configurations	Y	Y	Y	Y	Y
	Delays	N	Y	N	Y	N
	Bandwidth limit	N	N	N	N	Y
	Model protection systems in detail	Y	Y	Y	Y	Y
	Match expected site-specific equipment settings	Y	Y	Y	Y	Y
	Include any relevant non-linearities	Y	N	N	N	Y
	Sub-synchronous Oscillation (SSO)/SSCI mitigation and/or protection	Y	Y	N	N	N
Category	EMT Model Requirement	ERCOT	ISO-NE	MISO	Southern Company	AEMO
Model Usability Features	Represent the dynamics over the entire range of operating conditions.	Y	Y	N	N	N
	Have User-Accessible Control and Hardware Options	Y	Y	Y	N	Y
	Provide Diagnostic flags	Y	Y	Y	Y	N
	The model must operate accurately at a 10 μ s time step or greater.	Y	Y	Y	Y	Y
	Dynamic simulation solution parameters	N	N	N	N	Y
	Initialization	Y	Y	Y	Y	Y
	Allow Scaling of Active Power Capacity	Y	Y	N	N	Y
	Numerically stable	N	N	N	N	Y
	Include a test case	Y	N	Y	Y	N
	PSCAD simulation sets	Y	N	N	N	Y
	PSCAD messages	N	Y	N	Y	Y
	Documentation Requirements	Y	Y	Y	N	Y
	Model versioning	Y	Y	N	N	Y
	Updates to account for future versions of simulation tools	N	N	N	N	Y

Emerging Modelling Requirements - Large Demand

- TSOs realize emerging large demand (electrolyzers, data centers, crypto mining etc) can have significant impact on system stability.
- Research ongoing, but in general, EMT load model is required if:
 - size > X MW and transmission connected (e.g. 200MW by Energinet)
 - converter-based and with active controls, fast protection
- Model must be site-specific, including all relevant control and protection systems and suitable for dynamic studies from Hz to kHz (<10).
- Model needs to be verified by applying relevant set-point changes and required disturbance test (short-circuit, voltage, frequency)

Grid Forming (GFM) IBR Integration Worldwide



■ Benefits

- provide grid strength and facilitate connection of IBR in weak grid areas
- provision of fast frequency and voltage response
- system restart
- island operation

Ref: [GFM around the world on UNIFI website](#)

NESO: EMT Model Test for GFM Plant (1/2)

- NESO Grid Code includes three additional EMT model simulation requirements for plants with GFM capability:
 - **Active ROCOF Response Power:** GFM plant's capability to 'naturally' (<5ms) respond rate of change of frequency by injecting/absorbing MW power.
 - **Active Phase Jump Power:** GFM plant's capability to 'naturally' (<5ms) respond to voltage phase angle change by injecting/absorbing MW power.
 - **Active Damping Power:** GFM plant's capability to 'naturally' (<5ms) respond to low frequency oscillations by injecting/absorbing MW power.

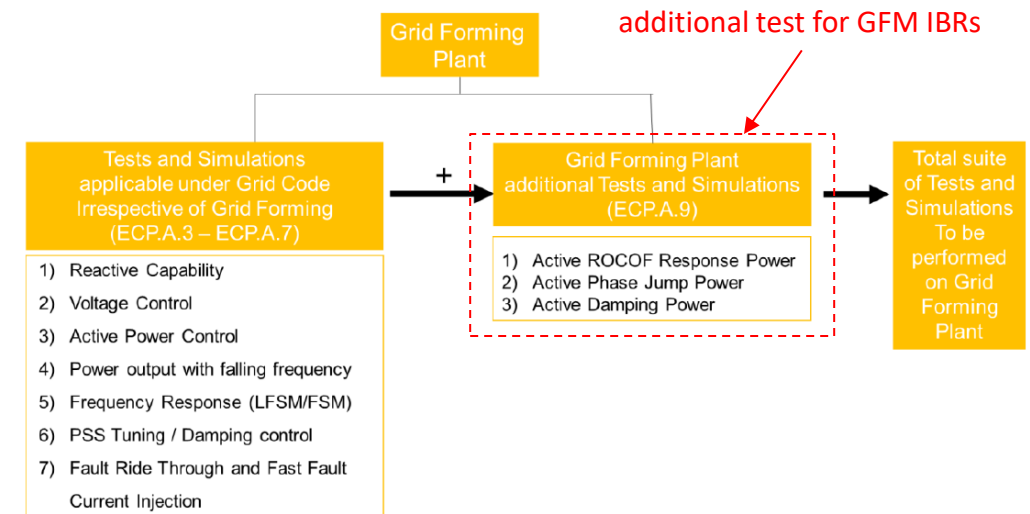


Diagram from "Great Britain Grid Forming Best Practice Guide", 2023.

NESO Example: EMT Model Test for GFM Plant (2/2)

- The figures show example results of conducting **Active ROCOF Response Power test** to a GFM BESS.
- System frequency was reduced with ROCOF of -2Hz/s for 1.5s to 47Hz and increase back to 50Hz.
- The GFM BESS with initial $P = 100\text{kW}$ respond to +/- ROCF by injecting or absorbing additional MW.

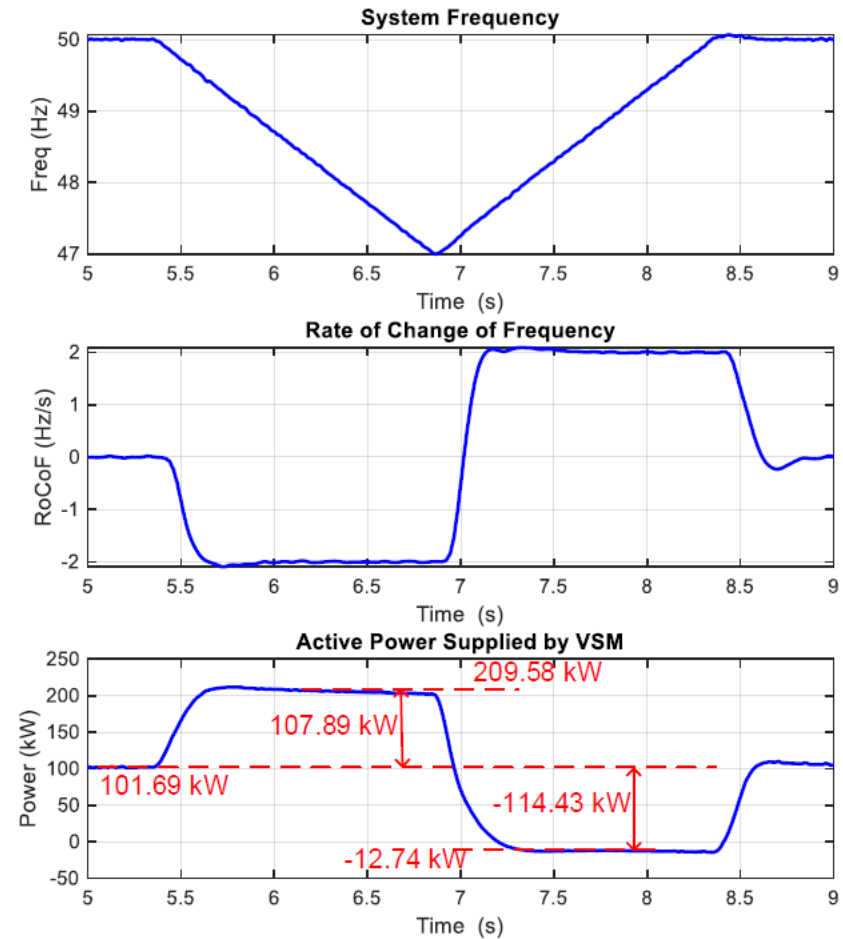
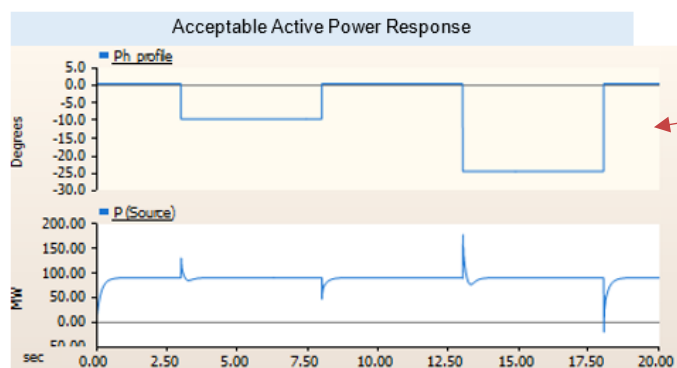


Diagram from "Great Britain Grid Forming Best Practice Guide", 2023.

ERCOT Example : EMT Model Test for AGS-ESR (1/2)

- The table shows ERCOT EMT model tests for IBRs and Energy Storage with Advanced Grid Support (AGS-ESR).
- Phase Angle Jump:
 - **IBRs:** find maximum angle jump magnitudes for IBRs to remain connected
 - **AGS-ESR:** apply angle step changes and check AGS-ESR MW respond to oppose the change



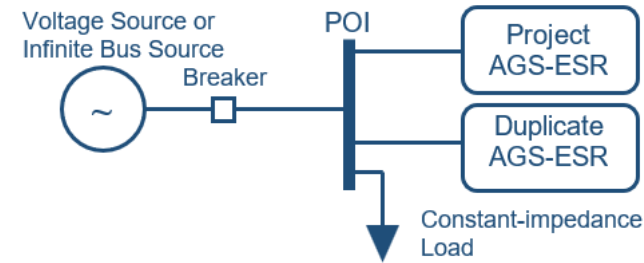
Technology	Tests
IBRs	1. Flat-start
	2. Small volt. disturbance
	3. Small frequency disturbance,
	4. Large voltage dist.: a. LVRT Test b. HVRT Test
	5. System Strength
	6. Phase Angle Jump
AGS-ESR (Energy storage with Advanced Grid Support)	1. Flat-start
	2. AGS Small volt. Disturbance
	3. AGS Frequency change and inertia response
	4. Large voltage dist.: a. LVRT Test b. HVRT Test
	5. AGS System Strength
	6. AGS Phase Angle Jump
	7. AGS Loss of synchronous machine

Diagram and Table from “Dynamic Working Group Procedure Manual Rev 24”, ERCOT, June 2025.

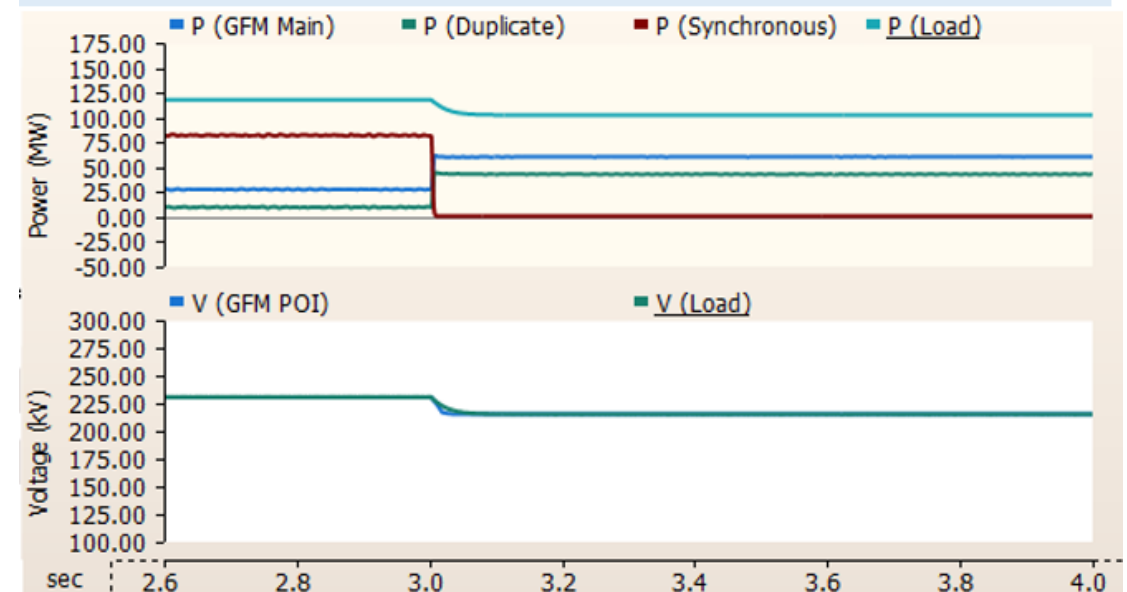
ERCOT Example: EMT Model Test for AGS-ESR (2/2)

- The purpose of the **Loss of SG test** is to demonstrate Energy Storage with Advanced Grid Support (AVS-ESR) to form voltage without synchronous generators.
- The test network includes **two identical AVS-ESR**, a load and a voltage source.
- Freq and voltage should settle to a stable operating point within 5s of voltage source disconnection.
- P and Q to change immediately to meet 90% demand within one cycle. Final P and Q value is based on the droop setting.

Test network for Loss of SG Test



Acceptable Response to a Loss of Synchronous Machine Test



Diagrams from “Dynamic Working Group Procedure Manual Rev 24”, ERCOT, June 2025.



Large-scale EMT Studies

Energinet Example: HVDC Commutation Failure in Denmark (1/2)

- A fault applied close to BJS substation, resulting in two 400kV transmission lines between BJS and ISH tripped and 400kV network separation.
- Western part of network became weak due to lost of SGs in eastern part. LCC HVDC SB and KO experienced commutation failure.
- A synchronous condenser near BJS tripped, further reduced strength of western part of network.
- SB HVDC tripped by SSO protection.

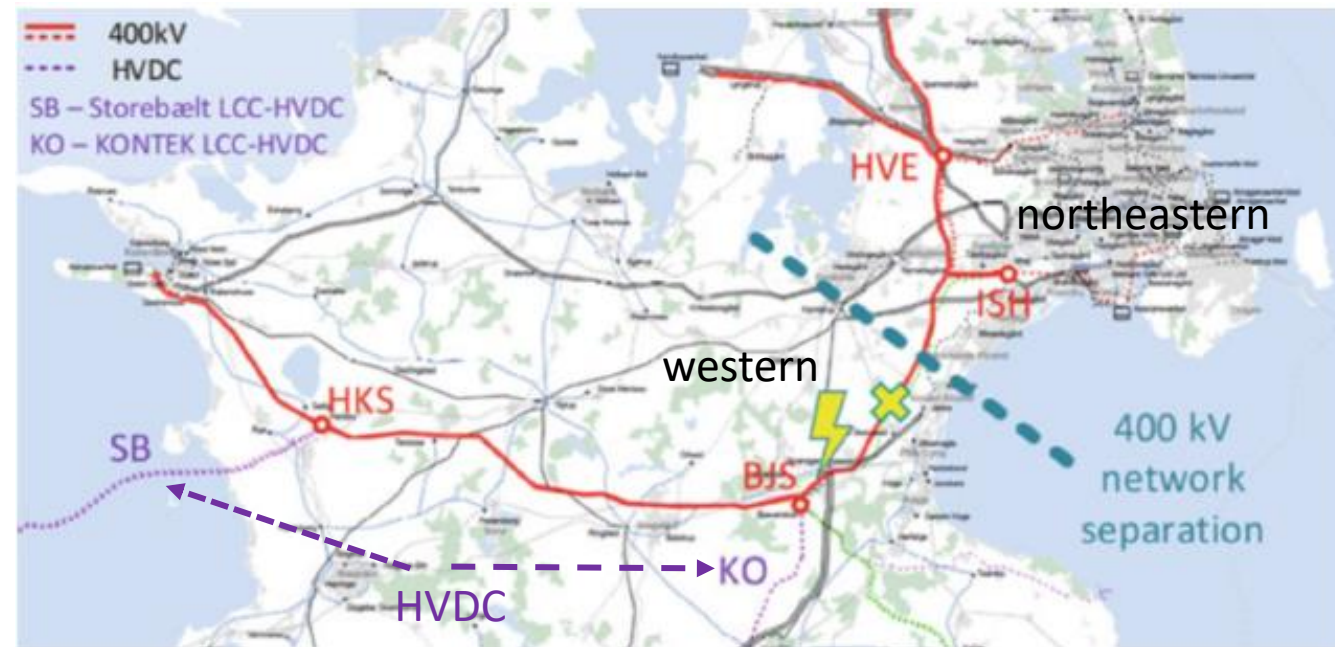


Diagram is from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

Energinet Example: HVDC Commutation Failure in Denmark (2/2)

- Studies were conducted in both PDT and EMT to investigate the fault event.
- PDT studies using vendor-specific PDT models did not reveal commutation failures, nor the SB-HVDC disconnection.
- EMT studies using vendor-specific models produce accurate results compared to filed measurements, including SSO behavior and the subsequent SB-HVDC disconnection.

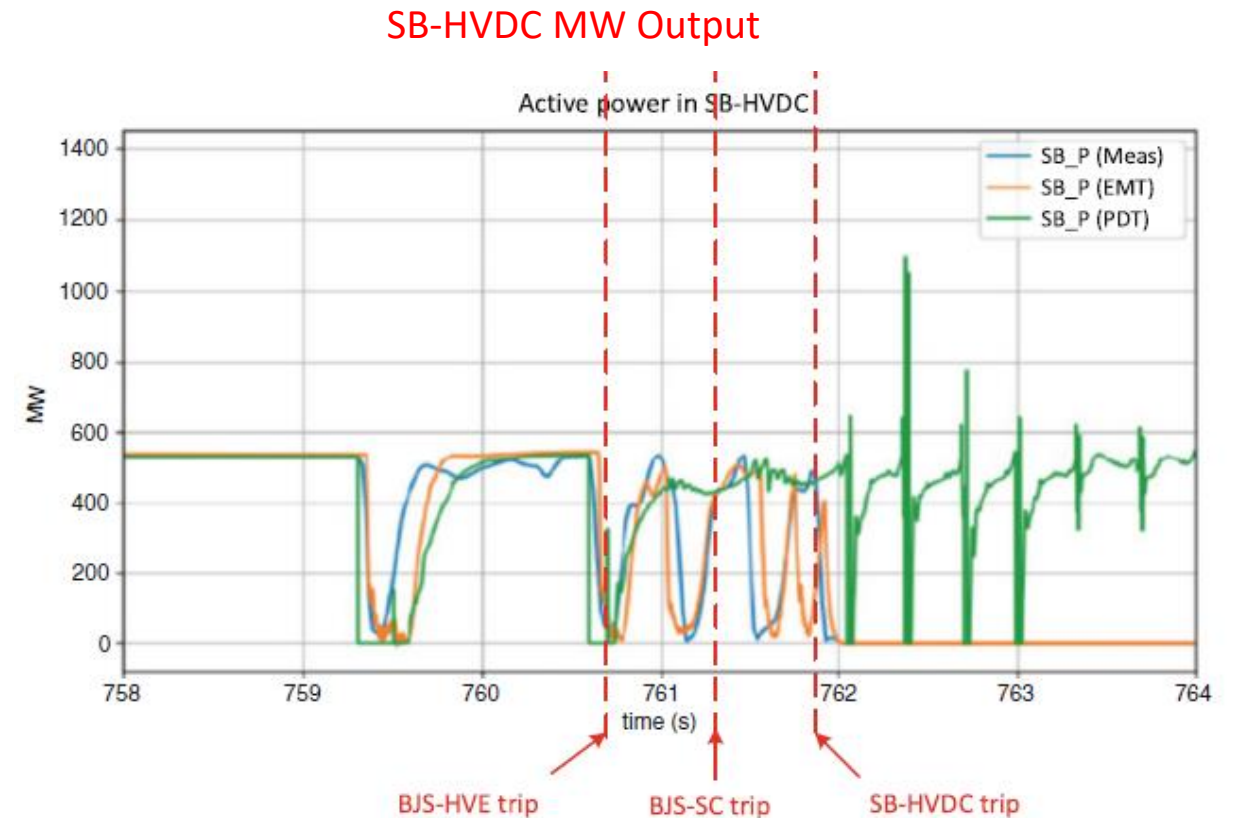


Diagram is from the CIGRE Green Book on "Power System Dynamic Modelling and Analysis in Evolving Networks" Eds. B. Badrzadeh & Z. Emin

Energinet Example: EMT Studies for Future Network Planning (1/2)

- Upgrading two 400kV lines in 2028 will temporality lead to **400kV network separated in two parts (western and northeastern)**.
- Energinet identified the **strength of western part will be greatly reduced** but still with **very high penetration of IBRs**.
- Comprehensive studies were conducted to investigate potential dynamic behaviors of the islanded western part under critical contingency conditions in EMT.

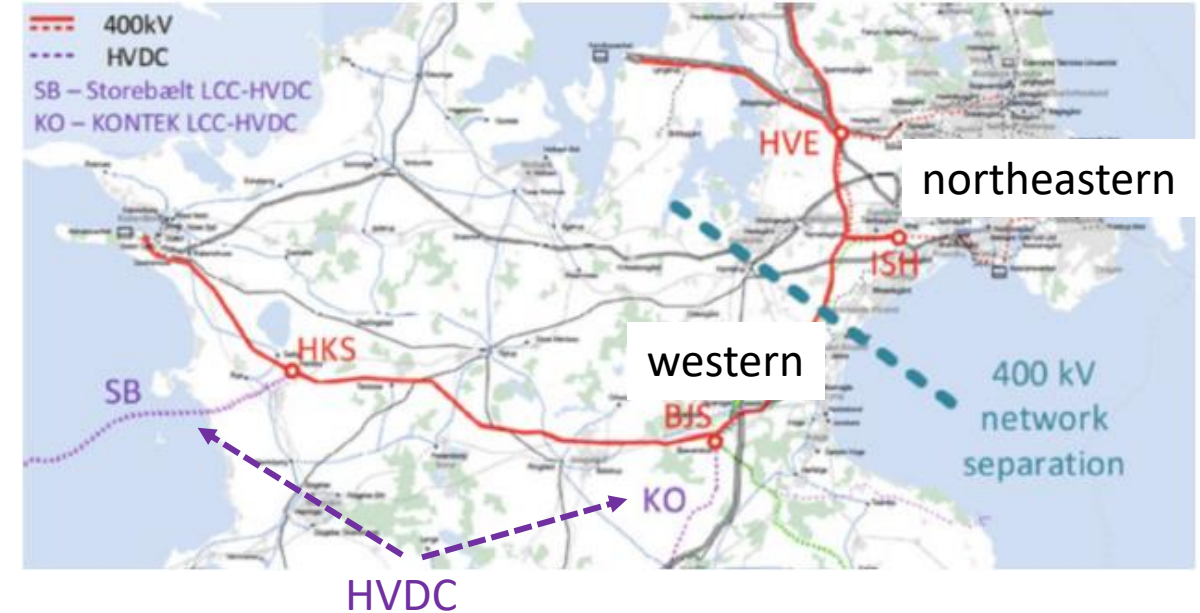
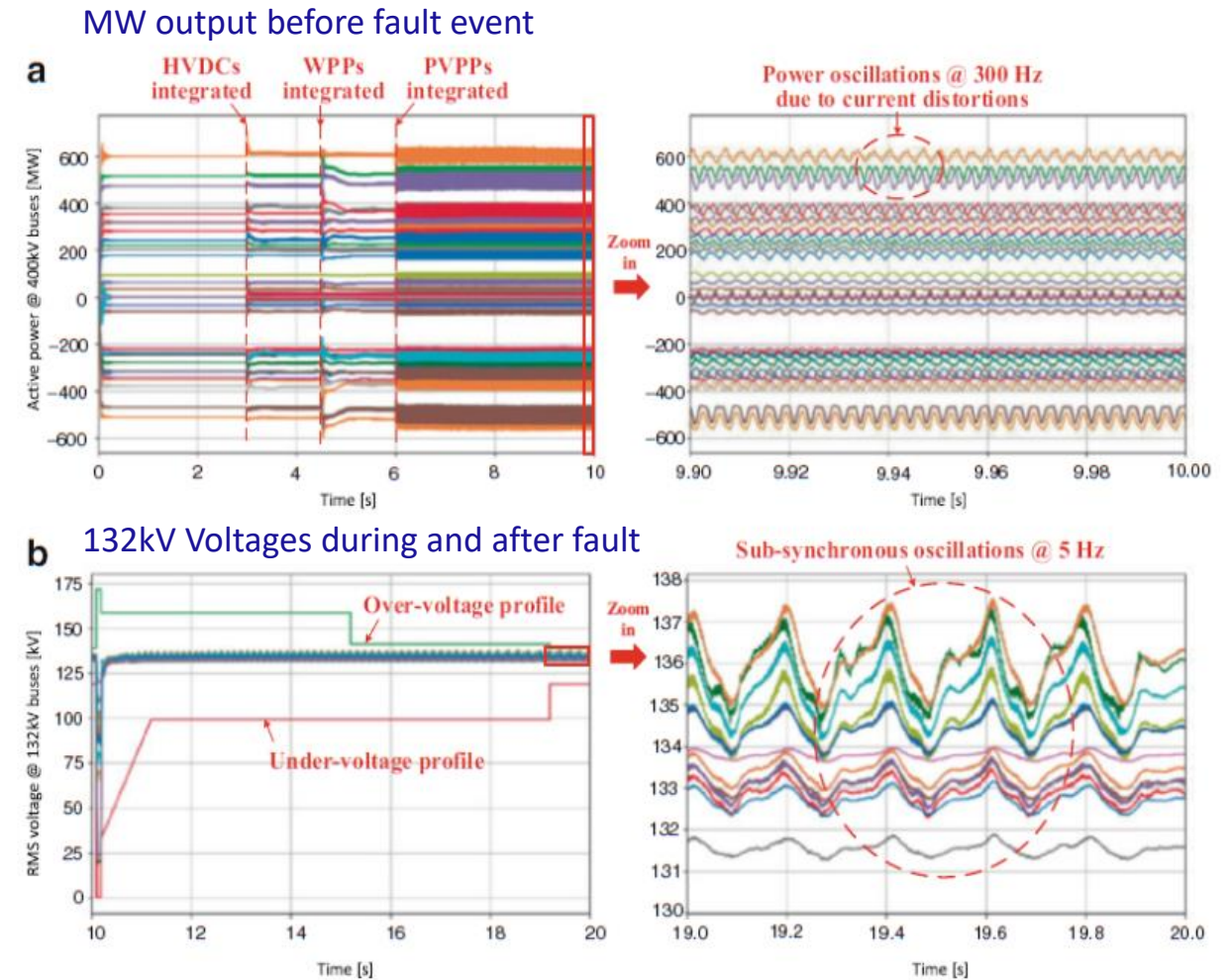


Diagram is from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

Energinet Example: EMT Studies for Future Network Planning (2/2)

- Figure shows EMT results of a three-phase fault at a 132kV transmission line.
- Figure (a) shows **large power oscillations around 300Hz** propagating through transmission networks due to current distortion before the fault.
- Figure (b) shows **post-fault SSO around 5Hz** at 132kV buses near PV plants.
- PDT results did not show any oscillations.



Graphs from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

Fingrid Example: EMT Studies for Outage Planning (1/2)

- Fingrid began carrying out large-scale EMT studies for outage planning since 2023.
- Several critical planned outages at 400kV were taken in the area shown in the figure in summer 2023.
- The study area has around 80 wind generation.
- The EMT model included the whole 400kV network, but for 110kV network only the generators locally and in the neighboring area.

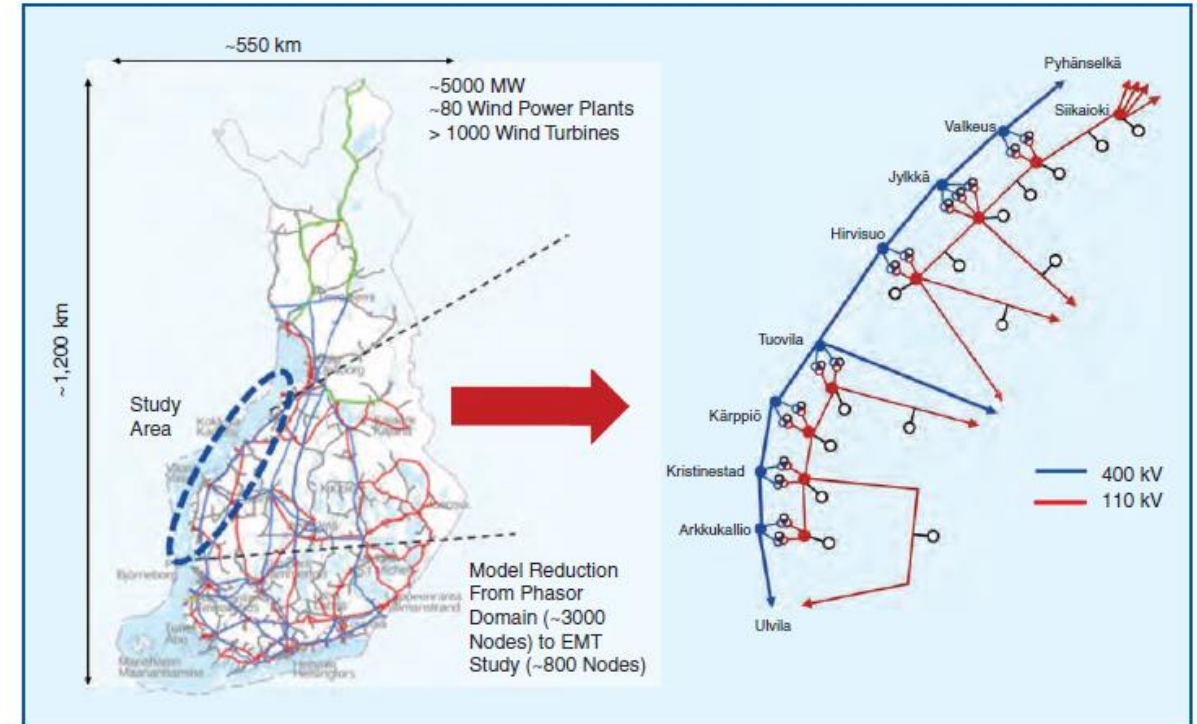


Figure from “Large-Scale EMT Model Examples for High-Share IBR Systems: Using Large-Scale EMT Models for Operations and Planning Power Systems With a High Share of IBRs”, IEEE Power and Energy Magazine, Vol 23, Issue 4, 2025.

Fingrid Example: EMT Studies for Outage Planning (2/2)

- The studies showed the total output from the windfarms in the study area **need to be curtailed below 2GW** from the total installed capacity (4GW) during the planned outages. Otherwise, **5-6 Hz large voltage oscillations** would happen leading to tripping of some windfarms.
- Some mitigations were also identified in the studies, such as **retuning some of the power plant controllers and setting older plants to constant Q control**. This would reduce the amount of curtailment required to 1.6GW.
- In another planned outage case, Fingrid also identified the need to curtail wind power by 1.3GW out of 1.5GW based on another large-scale EMT studies. Mitigations including **network topology change** was identified in the EMT studies and implemented in real environment to reduce the amount of curtailment needed.
- Fingrid's large-scale EMT model includes more than 2060 buses and 100 IBRs models.

ERCOT Example: Large-scale EMT studies for Weak Grid Region

- ERCOT have carried out large-scale EMT studies for weak areas since 2018.
- In 2020, EMT studies were carried out for west Texas 345kV network (Panhandle and nearby areas) with very high penetration of IBRs (>10 GW).
- The EMT model included more than 400 buses and 42 parallel models of 46 IBRs projects.
- It took 2.5 hrs to run one contingency case.

Figure from “2020 Panhandle Regional Stability Study”, ERCOT, August 2020.

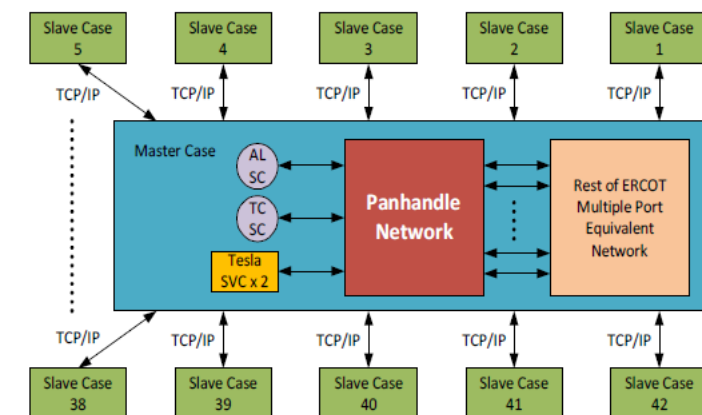
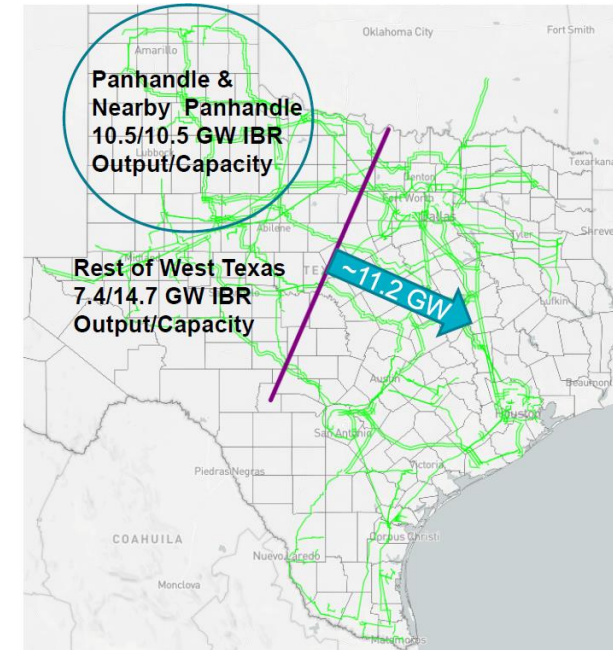


Diagram from “ERCOT’s Experience Integrating High Shares of IBR”, ERCOT, June 2021.

AEMO Example: Cloud-based EMT Simulation of Large Network

- AEMO developed **cloud-based** set up including AEMO's EMT electricity network model.
- Users can test their EMT models by connecting to the full network model **using dedicated cloud-based virtual machines**.
- Users only **have visibility of their own models and up to the connection point**.
- This allows the users to test their EMT models more comprehensively with better tuned parameters.

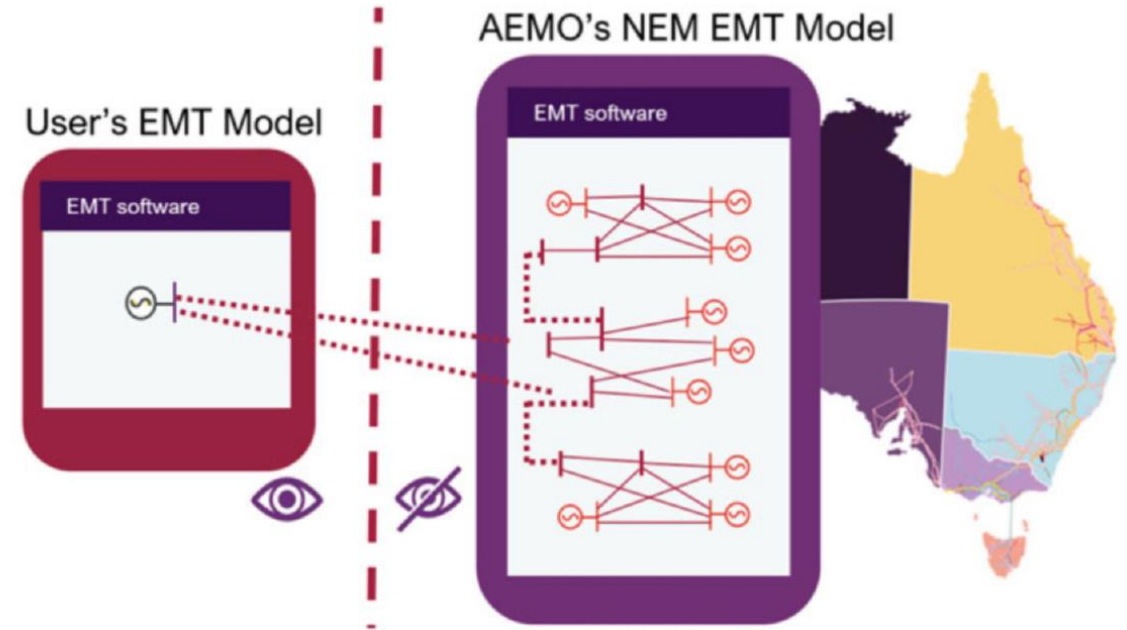


Diagram is from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

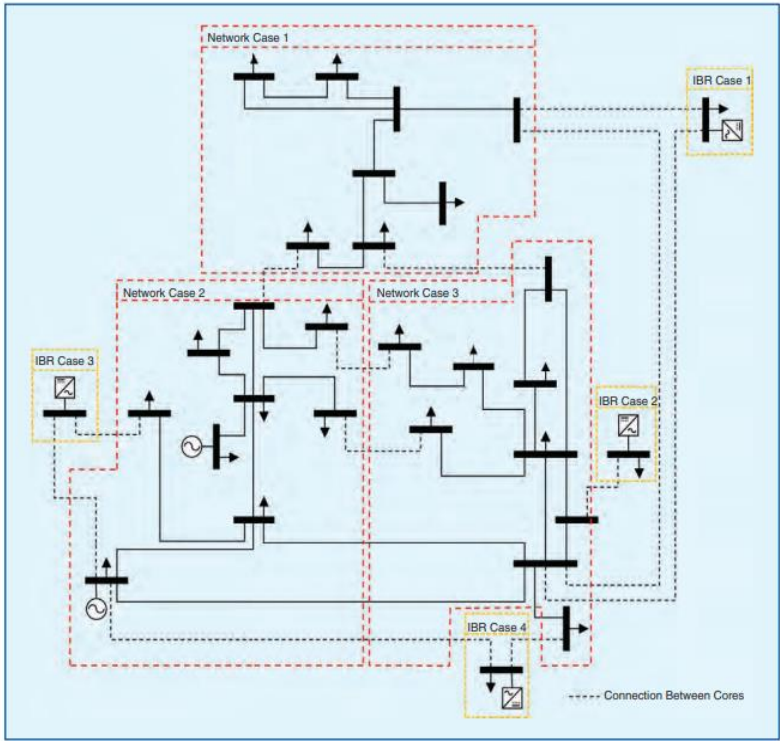
Reducing Simulation Time for Large-scale EMT Studies

- One key challenge in carrying out large-scale network EMT studies is long simulation time.
 - experience shows simulation time of less than 1 minute can take 24 hours
- Several factors affecting simulation time:
 - 1. software and hardware used
 - 2. simulation time steps, which often cannot be improved as application dependent
 - 3. model components that affect the matrix size, i.e. IBR switching elements and number of network buses
- Improvements to Factor (1) and (3) can be done to further reduce the total simulation time.

Large-scale EMT Model Structure for Efficient Simulation

- Figure shows EMT model structure for efficient EMT simulations.
- To reduce simulation time, IBR plant models are **run in separate parallel processors**.
- Similarly, network models are divided into smaller parts and run in separate parallel processors.
- Table shows simulation time improvement from TSOs with such EMT model structure.

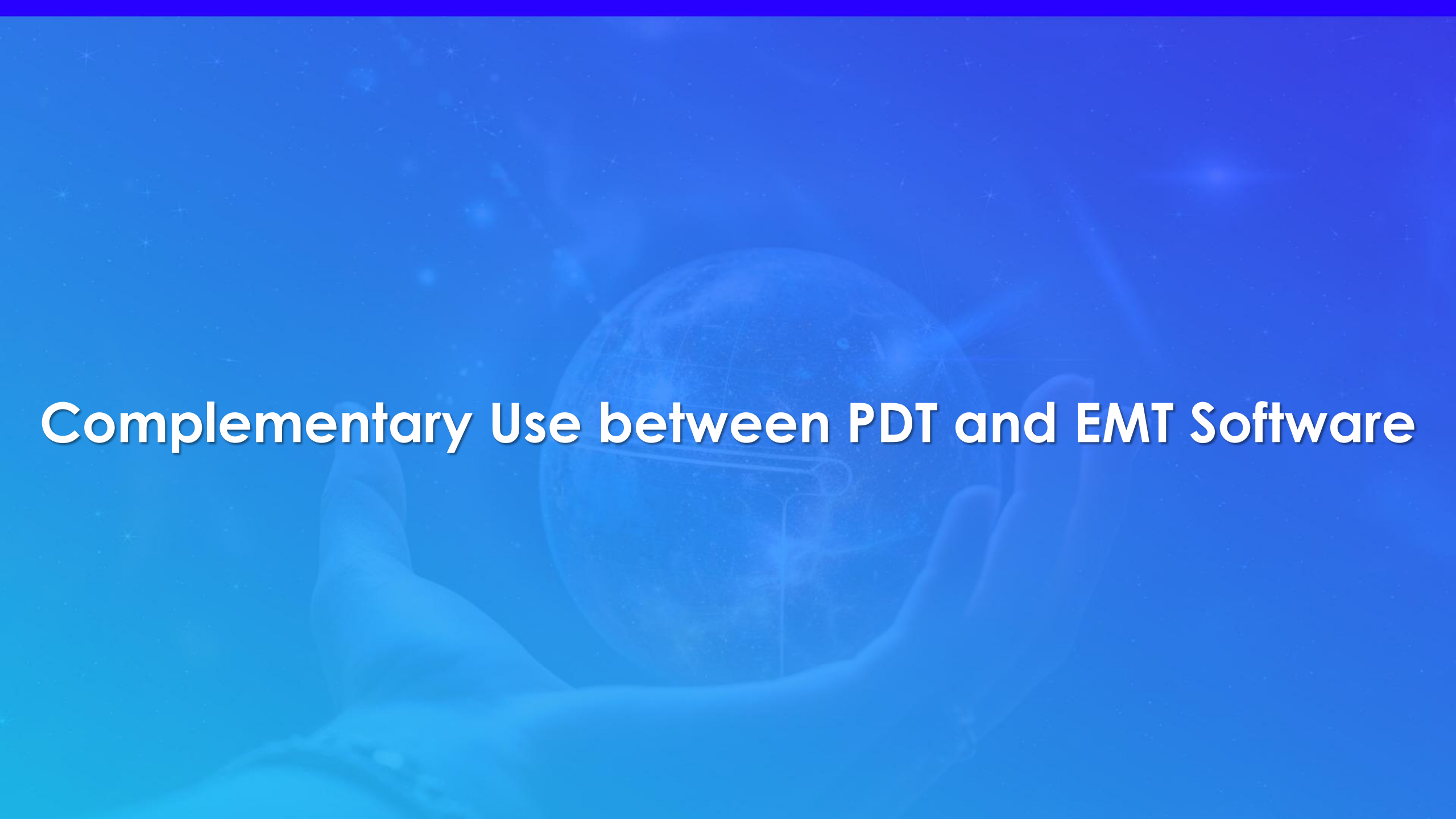
dividing large network model into smaller parts and run on separate computer cores



real time taken for running one large-scale EMT case

Country/ Region	Approximate Size (Number of Busbars)	Number of Parallel Cases	Simulation Duration (s)	Real Time for Simulation (Hours:Minutes)	Network Portion Modeled
Australia	3,000	240	30	3:15	Entire eastern mainland grid
Denmark	800	80	30	1:30	Two models: the eastern Denmark grid and the western Denmark grid
Finland	800	12	15	2:30	Areas of interest only (e.g., west coast area)
Texas, USA	400	43	30	2:00	Areas of interest only (e.g., South Texas)

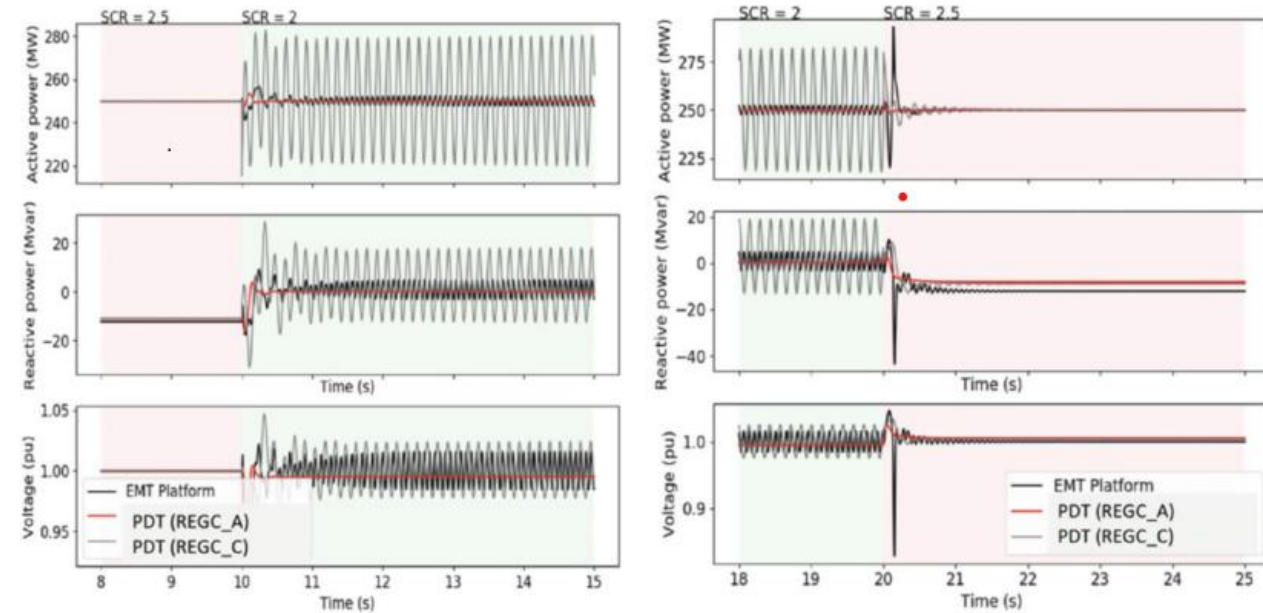
Table and Figure from “Large-Scale EMT Model Examples for High-Share IBR Systems: Using Large-Scale EMT Models for Operations and Planning Power Systems With a High Share of IBRs”, IEEE Power and Energy Magazine, Vol 23, Issue 4, 2025.



Complementary Use between PDT and EMT Software

PDT Generic Renewable Generation Model Improvements

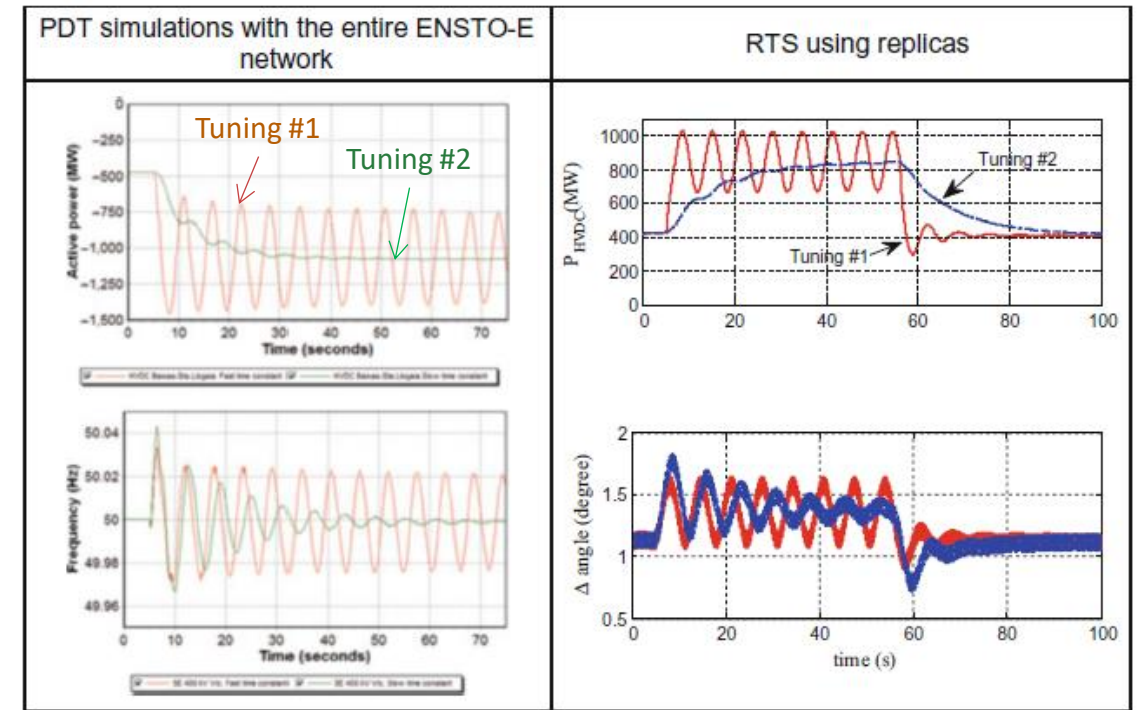
- The exact performance of PLL and fast control loops of IBRs can only be accurately modeled and simulated in EMT domain.
- However, recent modeling efforts have produced **enhanced generic PDT renewable energy generator model (REGC_C)** including an “approximate” representation of PLL and inner current control loop dynamics.
- As shown in the figure, REGC_C model allows **oscillatory behavior of IBRs under low system strength conditions to be observed** to some extends in PDT.
- Such PDT model improvement allows TSOs using PDT software as a screening tool.



Graphs from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

Example: Inter-Area Oscillations in Continental Europe

- An incident in 2016 caused the power system frequency in the **Iberian Peninsula and Turkey to oscillate against central and eastern Europe** with a frequency of 0.15 Hz.
- The figure compares PDT and real-time EMT simulation results with **two HVDC AV emulation control tuning #1 & #2**.
- Similar results were achieved due to PDT model was tuned based on the **exact function implemented in HVDC replicas**.
- The example shows the complementary use of PDT and EMT simulation tools.



Graphs from the CIGRE Green Book on “Power System Dynamic Modelling and Analysis in Evolving Networks” Eds. B. Badrzadeh & Z. Emin

Conclusions

- Since late 2010, TSOs have started building EMT study capabilities in-house mainly due to increasing awareness of new types of stability issues triggered by IBRs.
- Key activities involved defining EMT modelling requirements in grid codes, expanding EMT study team through training and recruitments, upgrading/improving IT & computing infrastructure and carrying out large-scale systemwide studies.
- Challenges were encountered such as gathering EMT models from existing plants, model portability, scarcity of EMT experts in the market, local vs cloud-based computing solutions, long simulation times when running large-scale EMT studies.
- TSOs are continuously improving EMT study capabilities and working on defining EMT modelling requirements for large loads and devices with grid-forming capability in grid codes, building dedicated EMT model management team, identifying use cases for real-time EMT applications etc.
- Results from large-scale EMT studies carried out by TSOs have proven to be valuable and have direct impact on network operation and planning to avoid system instabilities.
- TSOs expect complementary uses between PDT and EMT software.

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